

NexusRAG: Corrective Hybrid Retrieval with Sentence-Level Faithfulness Verification for Scientific Claim Evidence

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Abstract

Retrieval-augmented generation (RAG) is increasingly used to answer questions over scientific papers, but two practical problems undermine trust: retrieval quality is rarely measured against ground truth, and “verification” is often a citation-formatting check rather than a test of whether a source actually supports a claim. We present NexusRAG, a fully local RAG system, and a reproducible, CPU-only evaluation of it on SciFact, a scientific claim-verification benchmark. We report an additive ablation of every retrieval component (BM25, dense MiniLM embeddings, reciprocal rank fusion, query-adaptive fusion weights, keyword boosting, cross-encoder reranking, and MMR diversification) over 300 claims and a 5183-document corpus, with bootstrap confidence intervals and paired randomization tests. Dense retrieval alone reaches 0.648 nDCG@10; the full pipeline reaches 0.685. We then replace the system’s citation-index “verifier” with a natural-language-inference grounding model that localizes supporting evidence at the sentence level, and evaluate it against SciFact’s gold rationales: it attains 0.370 F1 at evidence-sentence selection and 0.527 label accuracy, where the citation-index check scores zero because it cannot inspect content. All numbers come from one command on a CPU-only laptop, no hosted API. Code and data: <https://github.com/urme-b/NexusRAG>.

1 Introduction

Scientific reading is a retrieval problem before it is a generation problem: the hard part of “what does the literature say about X ?” is finding the few passages that actually bear on X . Retrieval-augmented generation Lewis et al. (2020) couples a retriever with a language model so that answers are grounded in retrieved text, and a growing number of open-source tools apply this recipe to PDFs of papers. Two things, however, are usually missing from these tools.

First, the retriever is almost never measured. A typical project ships a hybrid retriever—dense embeddings plus BM25, fused and reranked—but no ground-truth relevance judgements, so it is impossible to say whether the hybrid actually helps, or which of its many moving parts matters. Second, verification is frequently cosmetic: systems confirm that a citation marker such as [2] points to a source that exists, never that the source entails the sentence it is attached to. A confidently wrong sentence with a well-formed citation passes.

This paper is a corrective study of one such system. We take NexusRAG, a local RAG application of exactly the kind described above, and ask two questions that its original form could not answer:

1. Which retrieval components earn their place? We build an additive ablation ladder and evaluate it on SciFact Wadden et al. (2020) using the BEIR Thakur et al. (2021) relevance

judgements, reporting nDCG@10, Recall@ k , MRR, and MAP with bootstrap confidence intervals and paired randomization tests.

2. Can verification be made real without sacrificing locality? We replace the citation-index check with a cross-encoder NLI model He et al. (2021) that scores, for each answer sentence, whether any source sentence entails it. We evaluate this grounding signal against SciFact’s sentence-level gold rationales.

Our contribution is not a new retrieval algorithm. It is (i) an honest, component-wise measurement of a popular RAG recipe on a scientific benchmark, (ii) a drop-in faithfulness verifier that inspects content rather than citation syntax, and (iii) a reproducibility artifact: every number in this paper is generated by `make eval` and `make faithfulness` on a CPU-only laptop, with a vendored data subset so the pipeline runs offline.

2 Related Work

Hybrid retrieval. Dense retrievers Karpukhin et al. (2020); Reimers and Gurevych (2019) and lexical scoring Robertson and Zaragoza (2009) have complementary failure modes, and fusing their ranked lists with reciprocal rank fusion (RRF) Cormack et al. (2009) is a strong, training-free baseline. Cross-encoder reranking Nogueira and Cho (2019) re-scores a shortlist with full query-document attention, and MMR Carbonell and Goldstein (1998) trades relevance for diversity. NexusRAG combines all of these; we measure each in isolation.

Corrective and self-reflective RAG. CRAG Yan et al. (2024) grades retrieved passages and triggers corrective actions (e.g. re-querying) when retrieval looks poor, and Self-RAG Asai et al. (2024) trains a model to emit reflection tokens that critique its own retrieval and generation. NexusRAG’s corrective loop follows the CRAG pattern; in this study we isolate its retrieval-side effect with a deterministic, LLM-free confidence gate so the result is reproducible without a generation model.

Faithfulness evaluation. RAGAS Es et al. (2024) decomposes an answer into claims and checks each against the retrieved context, typically with a large judge model. We pursue the same goal with a small local NLI cross-encoder and, because SciFact provides sentence-level gold evidence, evaluate the grounding signal directly rather than through another model’s opinion.

Benchmarks. BEIR Thakur et al. (2021) standardized zero-shot IR evaluation; SciFact Wadden et al. (2020) pairs scientific claims with sentence-level support/contradiction labels, letting us evaluate retrieval and grounding on the same data.

3 The NexusRAG System

NexusRAG ingests documents (PDF, DOCX, TXT, MD), chunks them, and indexes each chunk both as a MiniLM Wang et al. (2020) dense vector and in a BM25 index. At query time (the ladder of Figure 1) it fuses cosine search over normalized MiniLM embeddings with BM25 via reciprocal rank fusion (smoothing constant $k=60$), up-weighting the lexical list for short or technical queries and the dense list for long natural-language ones, and boosting chunks that contain query keywords. A MiniLM cross-encoder then reranks the shortlist, and an MMR pass removes near-duplicate passages. A synthesizer prompts a local Ollama model to answer using only the retrieved passages,

with inline citations. The original system “verified” that answer by confirming citation numbers were in range; we replace this with the grounding verifier of Section 5.3.

4 Experimental Setup

Data. We use SciFact via its BEIR packaging: a corpus of 5183 abstracts and 300 test claims with abstract-level relevance judgements. For the faithfulness experiment we use the original SciFact release, which adds sentence-level evidence labels; we tune the grounding threshold on the train split and report on the dev split (188 claims with evidence).

Retrieval metrics. We report nDCG@10, Recall@5/10/20, MRR, and MAP with binary relevance. Each system retrieves to depth 50. We compute 95% confidence intervals by bootstrapping over queries (10,000 resamples) and test each system against the full pipeline (the final configuration on the ladder) with a paired randomization test (10,000 permutations).

Ablation. The ladder is strictly additive: each row adds exactly one component to the row above, so the marginal effect of a component is the difference between consecutive rows.

Reproducibility. All retrieval results use exact (brute-force) cosine search rather than an approximate index, so they are deterministic. The dense model is all-MiniLM-L6-v2, the reranker is ms-marco-MiniLM-L-6-v2, and the NLI model is nli-deberta-v3-small. Everything runs on CPU; no API keys are required.

5 Results

5.1 Retrieval ablation

Table 1 and Figure 1 report the ladder. BM25 alone reaches 0.666 nDCG@10 and dense MiniLM 0.648; fusing them with RRF gives 0.670. The full pipeline (+ MMR (full)) reaches 0.685 nDCG@10 and 0.804 Recall@10, and the highest single configuration in the ladder is + Rerank (cross-enc) at 0.687. The table’s last column reports each row’s p -value against the full pipeline—a paired randomization test fixed a priori, not against a post-hoc winner: the full pipeline improves over dense-only retrieval ($p=0.031$), while the refinements between hybrid fusion and the full pipeline shift the mean within overlapping confidence intervals. We report uncorrected p -values; under a Holm correction for this family of six comparisons even the dense contrast is borderline, so we read it as suggestive evidence that fusing lexical and dense signals helps, and treat the later components as neutral at this sample size rather than as significant improvements. Values are means over 300 queries with 95% bootstrap intervals.

5.2 Generalization to a second corpus

To check that the component ordering is not an artifact of SciFact, we repeat the core retrieval ladder on NFCorpus Thakur et al. (2021), a biomedical IR benchmark (323 test queries, 3633 documents). Table 2 shows the same pattern, more sharply: the full core system reaches 0.344 nDCG@10 and significantly outperforms BM25 (0.311), dense retrieval (0.319), and even plain RRF fusion ($p=0.010$ for RRF vs. the full core system). Hybrid fusion plus query-adaptive weighting helps on both corpora; the gains are statistically reliable here, where the larger query set affords more

Table 1: Additive retrieval ablation on scifact (300 claims, 5183 documents). Each row adds one component. Best column values in bold.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP
BM25	0.666	0.719	0.790	0.821	0.636	0.626
Dense (MiniLM)	0.648	0.738	0.790	0.844	0.611	0.604
Hybrid-RRF	0.670	0.736	0.806	0.875	0.638	0.629
+ Adaptive weights	0.686	0.764	0.834	0.875	0.648	0.639
+ Keyword boost	0.677	0.758	0.812	0.869	0.645	0.635
+ Rerank (cross-enc)	0.687	0.745	0.812	0.875	0.661	0.648
+ MMR (full)	0.685	0.748	0.804	0.870	0.661	0.648

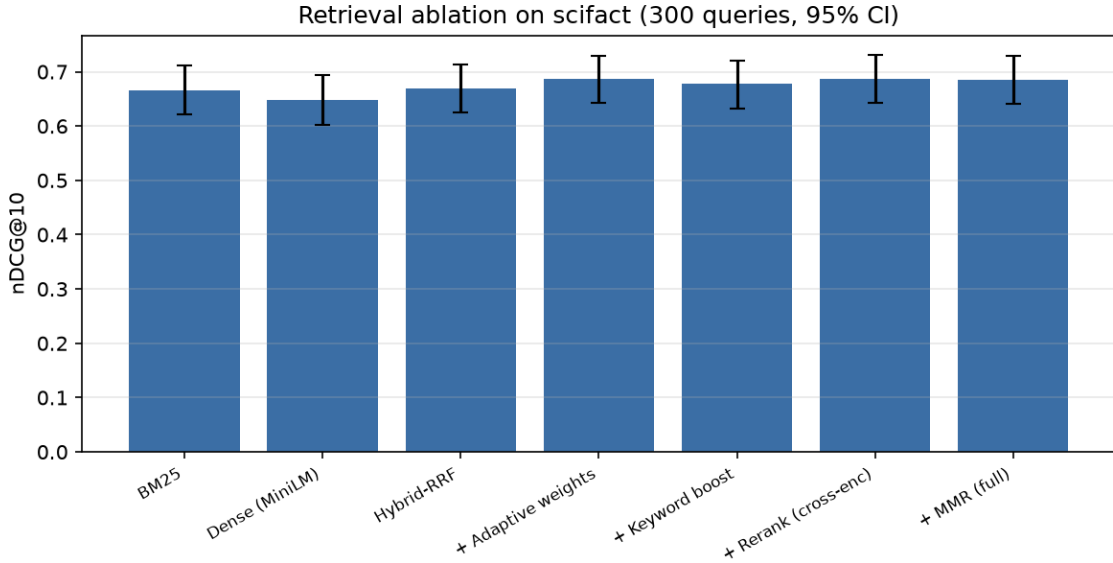


Figure 1: nDCG@10 across the ablation ladder with 95% bootstrap confidence intervals.

power. (We omit cross-encoder reranking and MMR on NFCorpus: its long full-text documents make the per-query rerank pass dominate runtime without changing the ordering.)

5.3 Faithfulness verification

We evaluate the grounding verifier as an evidence-sentence selector: for a claim and a cited abstract, it scores every abstract sentence by NLI entailment or contradiction against the claim and selects sentences above a threshold ($\tau=0.6$, tuned on the training split). Against SciFact’s gold rationales on the dev split (188 claims) it achieves precision 0.349, recall 0.393, and 0.370 F1, and predicts the support/contradict label with 0.527 accuracy. This is a zero-shot number from an off-the-shelf NLI model with no SciFact training; we do not present it as state of the art. The original citation-index check scores zero here by construction—it never reads the sentences—so the same pipeline can carry a content-inspecting verifier for the cost of one small local model, with a usable signal to gate the corrective loop.

Table 2: Core retrieval ladder on NFCorpus (323 queries, 3633 documents). Same ordering as SciFact.

System	nDCG@10	R@5	R@10	R@20	MRR	MAP
BM25	0.311	0.120	0.150	0.172	0.518	0.136
Dense (MiniLM)	0.319	0.120	0.159	0.189	0.510	0.136
Hybrid-RRF	0.334	0.127	0.162	0.202	0.544	0.152
+ Adaptive weights	0.341	0.132	0.164	0.207	0.554	0.159
+ Keyword boost	0.344	0.133	0.167	0.206	0.553	0.159

6 Discussion

Beyond nDCG, the benefit of combining lexical and dense evidence shows up in coverage: Recall@20 rises from 0.821 (BM25) and 0.844 (dense) to 0.875. The downstream refinements—adaptive weighting, keyword boosting, reranking, and MMR—move the mean within overlapping intervals at 300 queries; separating the components that reliably help from those that are neutral at this scale is the point a demo cannot make.

The faithfulness result reframes “verification”: a check on citation syntax offers no protection against a fluent, well-cited, wrong sentence, whereas an entailment check gives a real, if imperfect, signal and a place to gate the corrective loop.

7 Limitations

SciFact is biomedical-leaning; we mitigate the single-benchmark risk by replicating the retrieval ladder on NFCorpus (Section 5.2), but broader transfer is untested. The corrective loop in the main experiments uses a deterministic, LLM-free confidence gate for reproducibility, so its reported effect is a lower bound on an LLM-graded loop. We ship an end-to-end generation-faithfulness harness (nexusrag.eval.generation), but a smoke run with a 135M-parameter CPU model is uninformative (faithfulness ≈ 0.10 , no corrective gain); testing whether the loop reduces hallucination needs a capable generator. The NLI verifier operates at the sentence level, so multi-sentence reasoning chains are only partially captured, and generation quality itself is not scored here, by design.

8 Conclusion

We turned a typical local RAG application into a measurable one. The contribution is an honest, reproducible, component-wise evaluation on a scientific benchmark and a faithfulness verifier that inspects content instead of citation syntax. We hope the artifact is useful as a template for evaluating the many similar systems that currently ship without numbers.

Reproducibility. `pip install -e ".[eval]"` then `make eval` reproduces Table 1 and `make faithfulness` reproduces Section 5.3, both on CPU.

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